Finly: A Trading Agent That Knows What It Is Allowed to Decide

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Most trading agents begin with the question, "What should I trade?" Finly begins one step earlier: which parts of that decision should an AI be allowed to control? Its answer is a division of authority rather than another promise of machine foresight. The project contains three deliberately separate objects. **G4** is a retrospective research challenger, designed to make the strongest historical case against restraint. **tsmom_ensemble_vol** is the frozen production policy, a long-only SPY/BIL allocation intended to limit risk. The **options compiler** is neither strategy: it is the control boundary that can turn an already-permitted view into a defined-risk order intent. AI may assess bounded evidence, explain its judgment, or veto a proposal; deterministic code alone owns direction, horizon, structure, maximum loss, order fields, and the final PERMIT or NO_TRADE decision. Finly's thesis is simple: useful judgment does not automatically confer authority over capital.

G4 makes that thesis difficult rather than convenient. Every 21 sessions it assigned 50% to QQQ and divided the rest among the three strongest sector ETFs by their 252-to-126-session log return. From 2 January 2013 through 27 August 2026, its adjusted-close replay gained **967.11%**, or 18.97% annualized, against **580.82%** and 15.11% for SPY; its maximum drawdown was -28.99% against -33.72%. Yet the result did not survive Finly's promotion standard. Its Deflated Sharpe probability was 3.75% against a 95% requirement, the worst familywise-adjusted p-value was 0.3718 against 0.05, and both the static growth-control and authenticated source-overlap gates failed [3,4,7]. The research ledger conservatively records 113 heterogeneous attempts—including controls, invalidated runs, rejected suggestions, and reruns—not 113 independent strategies. The disclosed comparator for both equity studies is SPY buy-and-hold. G4 is a post-selected adjusted-close replay; the production figures below are a consumed theoretical adjusted-OHLC next-open ledger, not broker fills, options P&L, or evidence of future alpha.

The policy Finly actually froze is less theatrical and more useful. Every five sessions, tsmom_ensemble_vol averages the signs of SPY-minus-BIL trends over 21, 63, and 252 sessions, scales the resulting SPY weight toward a 10% volatility target, and leaves the balance in BIL [1,2]. In 415 next-open observations from 2 January 2025 through 28 August 2026, signals were formed at the adjusted close and filled at the following open. Charging five basis points on each absolute SPY and BIL traded-notional leg, the policy returned **+15.39%** with a **-5.45%** maximum drawdown; SPY returned **+33.52%**. At a deliberately punitive 25 basis points per leg, Finly remained positive at **+10.56%**. A separate one-basis-point small-account shadow—sell-first, $1 minimum orders, nine-decimal fractional quantities, and a fee proxy—turned **$300 into $351.88** [8]. This is not raw-return dominance. It is evidence that the production object does what it was designed to do: remain investable under friction while buying substantial drawdown protection.

The options layer applies the same principle at the point where mistakes become orders. A local model may supply a bounded score and rationale, but code derives the typed intent and compares every echoed field for exact equality. A shape, range, freshness, or equality failure becomes NO_TRADE; the model cannot enlarge loss, reverse direction, substitute a contract, or author an Alpaca payload. In the checked synthetic fixture, the compiler produced a one-contract SPY bear-put debit spread with an exact $366 maximum loss and $634 maximum gain. The challenge suite preserved direction through four source-removal tests and 32 perturbations, and an authenticated Alpaca paper-account interaction remained read-only [5,6,9]. These checks do not establish profitability. They establish something a P&L chart cannot: the boundary still holds when the model disagrees, when evidence is removed, and when a broker-shaped object is within reach.

Attempt 114 now converts that discipline into a prospective test. Before the first signal deadline, Finly publicly froze the protocol and the full 17-file settlement-and-inference runtime; a successful GitHub workflow and **23 unauthenticated GET checks** re-fetched the repository, commit, workflow, jobs, manifest, and every bound source file [10,11]. The fixed sample requires **254 consecutive pre-execution signal commitments** to obtain **252 settled returns**, with no skipping, backfill, replacement window, repeated confirmatory test, or optional stopping. The primary endpoint is the mean daily net log-return difference between the frozen production policy and SPY under the same five-basis-point cost rule. The public GitHub record verifies observed pre-deadline publication, not an independent cryptographic timestamp, provider origin, broker execution, or a performance result. That distinction is precisely the point. Finly does not ask judges to confuse a compelling backtest with knowledge. It asks them to recognize a trading agent built to know the difference—and a public protocol capable of earning the stronger claim later.

References. [1] Moskowitz, Ooi, and Pedersen, "Time Series Momentum," *JFE* 104(2), 2012, doi:10.1016/j.jfineco.2011.11.003. [2] Moreira and Muir, "Volatility-Managed Portfolios," *JF* 72(4), 2017, doi:10.1111/jofi.12513. [3] Bailey and López de Prado, "The Deflated Sharpe Ratio," *JPM* 40(5), 2014, doi:10.3905/jpm.2014.40.5.094. [4] White, "A Reality Check for Data Snooping," *Econometrica* 68(5), 2000, doi:10.1111/1468-0262.00152. [5] Alpaca, "Options Trading". [6] Alpaca, "Create an Order". [7] Finly, G4 robustness record. [8] Finly, equity execution-realism audit. [9] Finly, Alpaca MCP runtime audit. [10] Finly, Attempt 114 protocol and runtime manifest. [11] Finly, public pre-signal publication receipt.